

Welcome to
ECE 1965
Introduction to Electronics and Computer Engineering

Lecture 1: Introduction

Outline –Week 1

1. Intro
2. Course Overview
3. Electronics - Introduction
4. Basic concepts -Current, Voltage, Resistance, Conductance
5. Ohm's Law
6. Electronic Components

Course overview

What is electronics?

Electronics can be defined as a branch of physics and engineering that deals with the study of electron behavior, flow, and control under different conditions.

- **Key aspects of this course:**
  - **Electronics fundamentals –** Electronics/electric quantities – Voltage , Current, Resistance, Power
  - **Devices and components** - Involves components like resistors, capacitors, inductors, transformer etc.
  - **Electronic circuits:** Series and Parallel circuits and its analysis
  - **Magnetism and Electromagnetism –** Transformers, Motors
  - **Semiconductors -** Transistors, diodes, and some applications
  - **Digital electronics** – Binary number, Logic gates, and ICs
  - **Instrumentation and control fundamentals**
  - **Microprocessor/computer fundamentals**
  - **Electronics communications fundamentals**

Week #	Dates	Lecture: Module/Lesson	Lab Activities
Week 1	1/15/2025	Introduction: Course and syllabus, Basics of Electronics	Exp #1: Lab Equipment Introduction, Basic Circuit building
Week 2	1/22/2025	Ohms Law, Metric Convention	Exp #2: Resistors color coding, Ohm's Law
Week 3	1/29/2025	Resistors circuits: Series and Parallel Resistor Circuits	Exp #3: Series and Parallel Circuits
Week 4	2/5/2025	Capacitors	Exp #4: Multisim Introduction
Week 5	2/12/2025	Midterm 1	Exp #5: LED Soldering
Week 6	2/19/2025	Principles of Electro-magnetism and AC Circuits	Exp #6: RC Time Constants
Week 7	2/26/2025	Inductors and Transformers	Exp #7: Transformers
Week 8	3/5/2025	Spring Break	
Week 9	3/12/2025	Electric Machines: Motors and Relay Circuits	Exp #8: Relay Lab
Week 10	3/19/2025	Semiconductors, Diodes and Rectifier Circuits	Exp #9: DC Motor
Week 11	3/26/2025	Transistors: Amplifiers, Switches	Exp #10: Bridge Rectifier
Week 12	4/2/2025	Midterm 2	Exp #11: Bipolar Junction Transistor
Week 13	4/9/2025	Intro to Digital Electronics: Binary Numbers and Logic Gates	Exp #12: Logic Gates: Soldering Project: Solder a robot kit
Week 14	4/16/2025	Introduction to Instrumentation	Soldering Project: Solder a robot kit
Week 15	4/23/2025	Introduction to Computer and Electronics Communications	Solder project completion and evaluation
Week 16		Finals Week	

Course Objectives

1. Understand basic electrical/ electronics principles.
2. Know how to use electronic measuring tools and instruments.
3. Be able to read and build circuits from electrical and electronic diagrams.
4. Have knowledge of essential analog and digital electronic components and circuits.
5. Understand basic ideas of instrumentation, computer and communication systems.
6. Know how to use software to simulate electronic circuits - Multisim
7. Show soldering skills by putting together a robot car kit.

Assessment

- **Grading policy:**
  - Assignments and Quizzes: 20 %
  - Lab works: 25 %
  - Project: 10 %
  - Mid-term 1: 10%
  - Mid-term 2: 15%
  - Final Exam: 20%

Grading Scale

A—excellent (90 –100 %)
B—good (80 –89%)
C—acceptable (70 –79%)
D—poor but passing (60-69%)
F—failure (0 –59%)

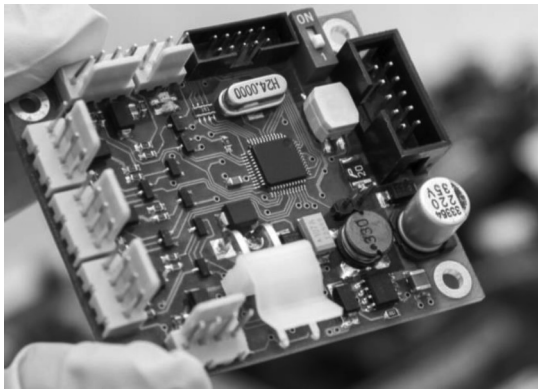
Secret for your Success

- Be Attentive: *Attention is all you need!*
- Get Organized :*Create a study schedule (At least 1-2 hours a week outside class); Manage your time wisely!*
- Stay Curious: *Look for connections between what you are learning and the real world.*
- Utilize Resources - *Take advantage of all the resources available to you—Study groups, online resources, library facility and textbooks.*

Can you name any modern gadget without any electronic component?

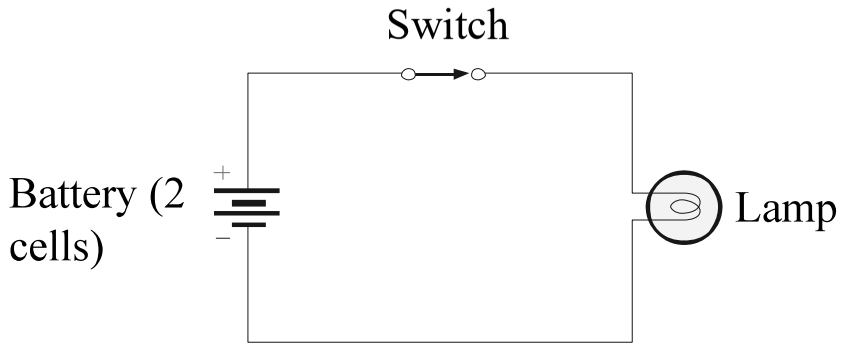
Electronics is the foundation of modern technology – Electronic components are found in almost all modern devices







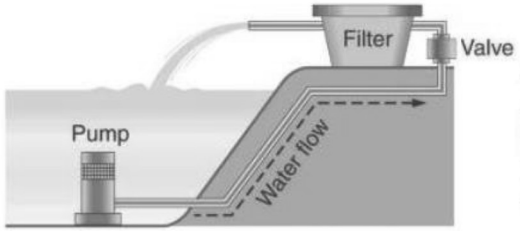
Electronics Fundamentals



A simple electric circuit

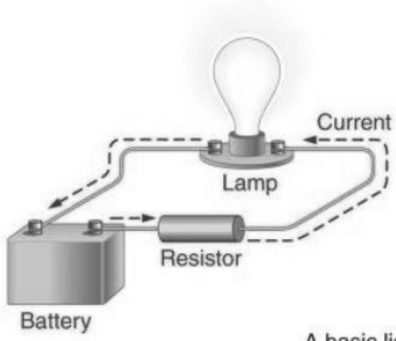
Electronics Fundamentals – Voltage, Current, Resistance

The Oldest Analogy in Electronics



The **pump** provides the **hydraulic force** that pushes **water** through the **pipe**. The rate of **water flow** depends on the pressure provided by the pump, and cannot exceed the capacity of the pipe. The **valve** is used to set the water flow to some desired rate.

A water garden filtration system



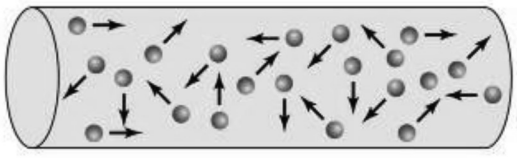
The **battery** provides the **electromotive force (EMF)** that pushes **charge** through the **wire**. The **current** depends on the EMF provided by the battery, and cannot exceed the capacity of the wire. The **resistor** is used to set the current to some desired rate.

A basic lighting circuit

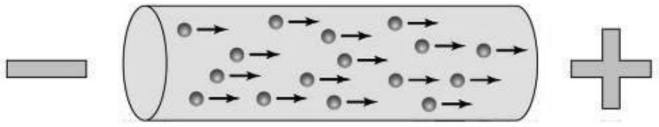
Voltage: The pressure pushing electricity.
Current: The flow of electricity through a circuit.
Resistance: Anything that slows down the flow of electricity.

Current

- Free electrons are available in conductors and semiconductive materials
- Free electron motion is random unless outside force is applied
- Electrons flow from negative to positive when a voltage is applied across a conductive or semiconductive material
- Current is **a directed flow** of charge
  - Electric current is the rate of flow of charge.



Random motion of free electrons in a materials



Applied Voltage

Electric Current: Flow of **electrons** through a **conductor** in a controlled fashion

Current

Current (I) is the net amount of charge (Q) that flows past a point in a unit of time (t).

- The defining equation is: $I = \frac{Q}{t}$

I = Current in Ampere (A)

Q = the charge of electrons in Coulombs (C)

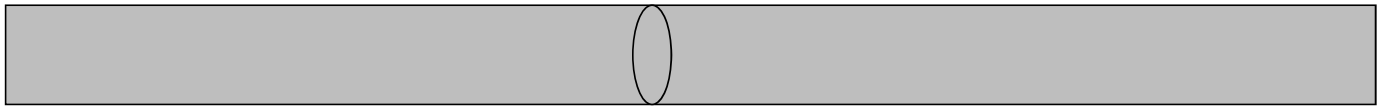
T = time in seconds (s)

Ampere (A): The unit of Current

Electric charge (Q) is measured in Coulombs.

1 Coulomb = Total charge possessed by 6.25×10^{18} electrons

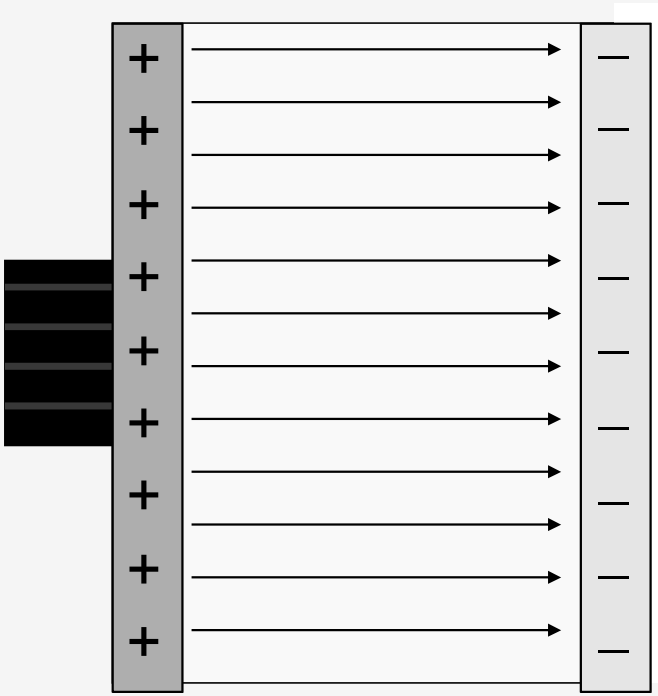
What is the charge of an electron?



What is the current if 2.0 C passes a point in 5.0 s?

I = 2.0/5.0 = 0.40A

Voltage



Force is required to move a charge against the electric field.

When force is applied over a distance, work is done. The work done in moving a charge against the electric field leads to the definition of voltage:

Voltage is the amount of energy per charge available to move electrons from one point to another in an electric circuit.

Voltage or electric potential difference or Electromotive force (EMF)

- The potential difference that drives current through a circuit. Think of it as the "pressure" pushing electrons.

Volt (V): Unit of Voltage

The defining equation for voltage is

$$V = \frac{W}{Q}$$

Where,

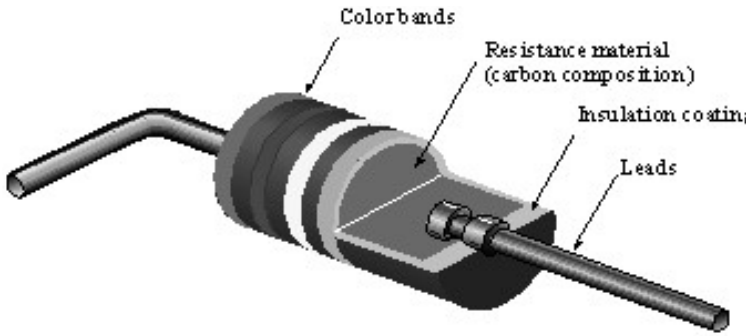
V is voltage (volts)
W is energy (joules)
Q is charge (coulombs)

One volt is the potential difference (voltage) between two points when one joule of energy is used to move one coulomb of charge from one point to the other.

Resistance

- Resistance – opposition to current
- Unit of Resistance – Ohm (Ω - Greek letter omega)
- Ohm – the **amount of resistance** that **limits current** to **one ampere when one volt is applied**
- One ohm (1 Ω) is the resistance exits when there is one ampere (1 A) is in a material with one volt (1 V) is applied across the material.

Components designed to have a specific amount of resistance are called resistors.



Resistance and Conductance

Conductance – **a measure of the ease** which current will pass through a component

$$G = \frac{1}{R}$$

where

G = *conductance*

R = *resistance*

Unit of Conductance – Siemens (S)
Old Unit of Conductance – mhos (upside down – Omega symbol)

Summary

Electrical Property	Units	Description
Charge (Q)	Coulombs (C)	1 C= 6.25 x 10^{18} electrons
Current (I)	Ampere (A)	The rate of flow of electrical charge 1 A = 1 C/s
Voltage (E or V)	Volts (V)	Electric Force
Resistance (R)	ohm (Ω)	The opposition to current.
Conductance	siemens (S)	The ability of a circuit to allow current

Announcements

- Lab 1 this week – Introduction to basic electronics equipment and Electronics components
- Watch the prelab videos before coming to the lab
- Quiz/Assignment 1 – Will be available on Canvas
- See you all at TB 120 (Friday)